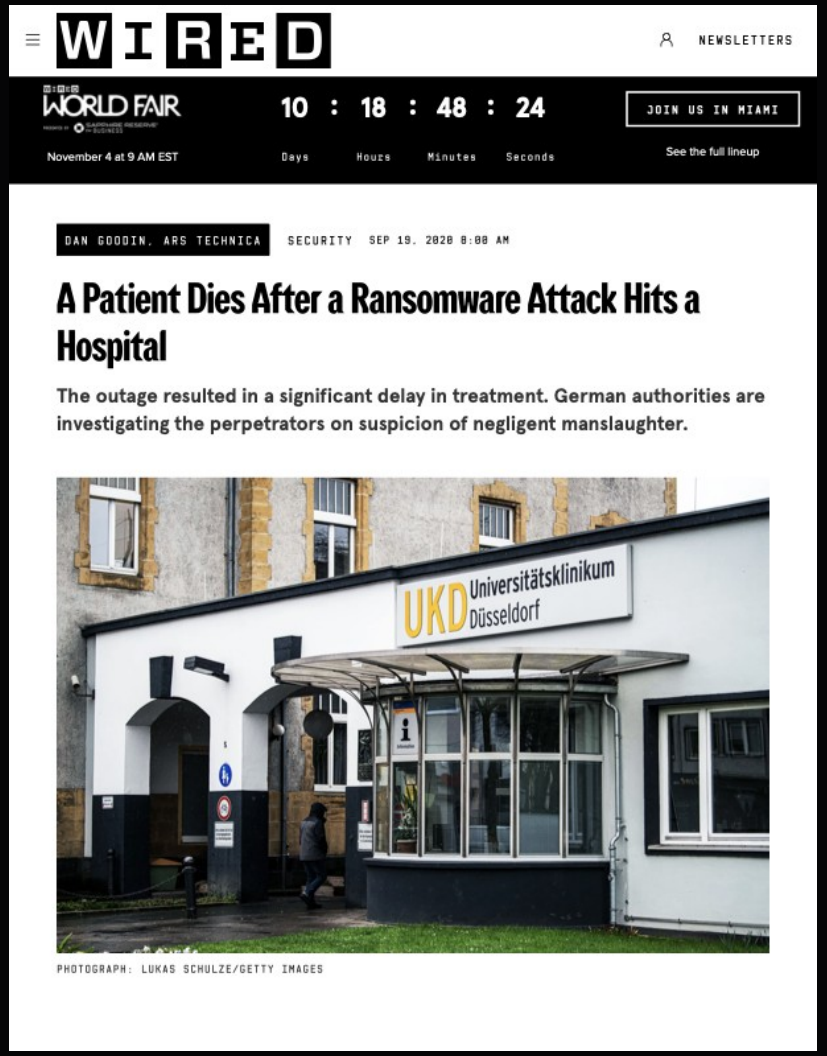


MIT Technology Review

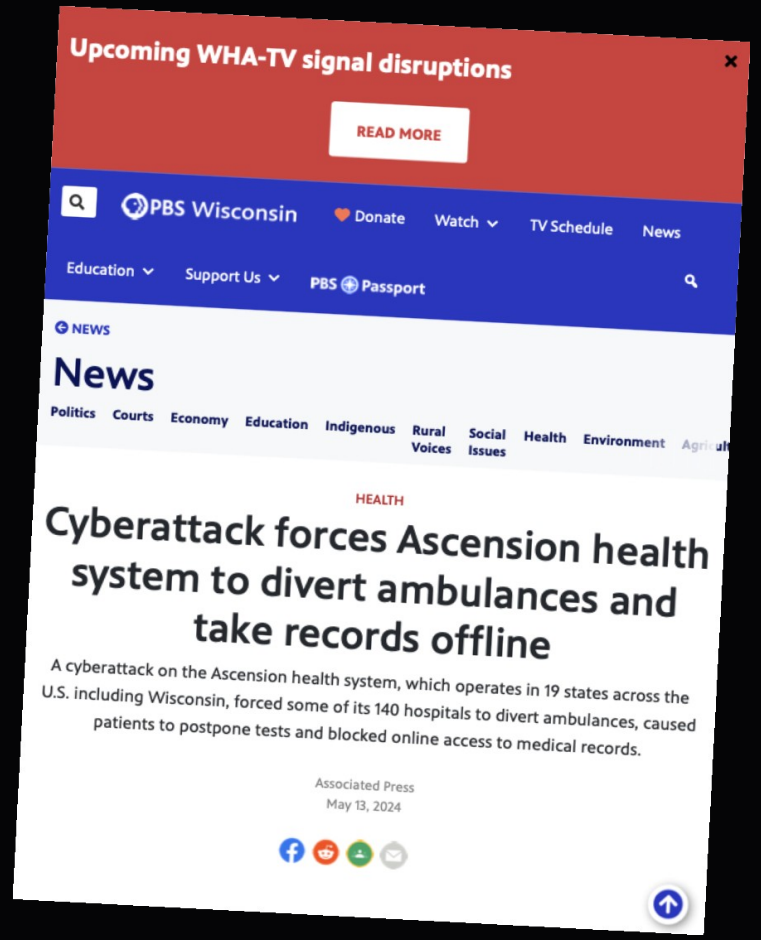
PATRICK HOWELL O'NEILL · SEP 18 · 2020

first documented ransomware death



WIRED

DAN GOODIN · ARS TECHNICA · SEP 19 · 2020



PBS Wisconsin

ASSOCIATED PRESS · MAY 9 · 2024

4 years later · same attack class

Different outlets. Four years apart. Same known vulnerability class · no tool to find, fix, and verify it at scale.

The device keeps reporting *"normal."* The patient keeps crashing.

WHAT THE SCREEN DISPLAYS

HR

72

SpO_2

98%

Green reads. Alarm silent. Monitor insists the patient is fine.

WHAT THE PATIENT ACTUALLY HAS

HR

32

SpO_2

76%

Fan stopped. Respiration flat. The alarm was suppressed by the attacker.

No doctor sees the difference. → ***That is why these attacks kill.***

Ransomware doesn't break the device — it *pivots through it.*

460

*US healthcare cyberattacks in **2025.** Up from 249 in 2023.*

FBI IC3 · 2025 ANNUAL REPORT

~53%

*Of connected medical devices carry a **known** critical vulnerability. Nobody writes the patch.*

CLAROTY · MEDIGATE · 2024 FLEET SURVEYS

One loop.
Visible *end to end.*

01	**Ground** *CVE-2019-6538*	Cite the real CVE from NVD. Not fabricated.	NVD
02	**Scan** *app.py*	SemGrep + Bandit run deterministic rules. Claude triages the real finding.	SEMGREP · BANDIT · CLAUDE
03	**Exploit** *POST /device/command*	Open-weight model writes the PoC payload. The device obeys.	PoC PAYLOAD
04	**Patch** *require X-Device-Token*	Factory Droid writes a real git diff. 41/41 unit tests green.	FACTORY DROID
05 · gate	**Human** *approve & deploy*	Mandatory sign-off. Tool-scoped, process boundary, identity-check.	REQUIRED
06	**Verify** *re-run the same bytes*	Replay the exact exploit. Hard-asserted status code — no LLM in the loop.	BYTE-FOR-BYTE REPLAY
07	**Restore** *healthy baseline*	Device back online. Receipt signed. Audit queue notified.	SIGNED

Find the hole. ***Prove the kill.*** *Close it. Show the receipt.*

The model shows its work — and *catches itself.*

LIVE · PREVENT DASHBOARD



01 STREAMING REASONING

Scan agent reads app.py line-by-line, visibly cross-checks the auth decorator.

02 VISIBLE CONFIDENCE

72% on the real finding. 58% after it rejects its own first patch.

03 SELF-CORRECTION EVENT

"REJECTED my own patch — v1 broke the emergency override path."

04 AUTONOMOUS PATCH

Factory Droid produces a real git diff. 41 / 41 unit tests green.

We re-run the exact exploit. It must now *fail.*

BEFORE PATCH

200

→

AFTER PATCH

401

VERIFICATION · BYTE-FOR-BYTE REPLAY

Before	POST /device/command
	→ 200 OK
	fan stops · alarm silent · patient crashing
After	POST /device/command
	→ 401 Unauthorized
	fan running · alarm armed · patient breathing
Judged by	hard-asserted status code
	no LLM in the loop

Pass or fail. ***No middle ground.***

A receipt the FDA *already asks for.*

FDA §524B · AUDITABLE RECEIPT

Model VENT-SIM / Lab unit *#0001*

Finding	CWE-306 · CVE-2019-6538
Scanner	SemGrep 1.182.1 · Bandit 1.9.4
Patch author	Factory Droid · scoped CLI
Approved by	human@ops · identity-signed
Verify	same exploit → 401 after fix
SBOM	attached · 73 packages
Reproduce	git · diff + exploit payload sealed to this record

✓ *Sealed and queued to clinical engineering* · 2026-10-24 11:17 PDT

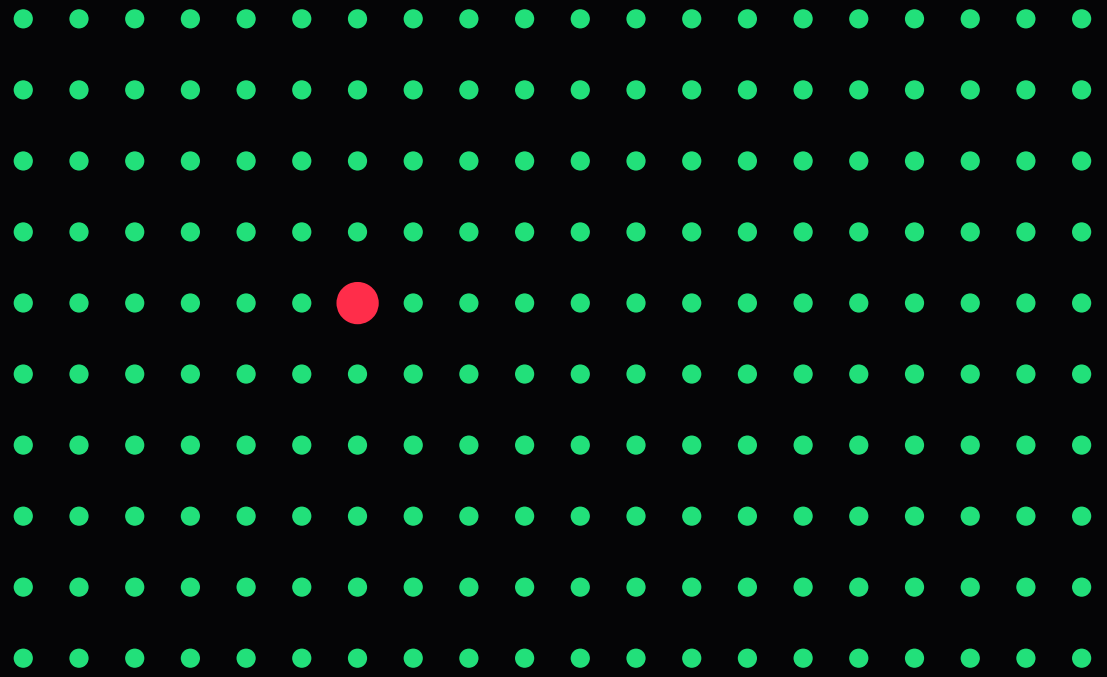
Patch one room.
The *heal-wave* rolls across the ward.

1 of

3,412

devices in a single hospital system

every one a parallel find → fix → verify job



• patched & verified ● one device PreVent is patching today

Make the devices *watch themselves.*

Find the hole. Prove the kill. Close it. Show the receipt.

01

Israel *Ogwu*

AGENTS · ORCHESTRATION

UMD · Anthropic Campus Ambassador · Claude Builder Club.

02

Christian *Ibarbia Ruiz*

SECURITY · INTEGRATION

UF · ex-Google Gmail · agentic full-stack.

03

Saketh *Poori*

HARDWARE · SIMULATION

14 hackathons · 8 awards · CV + hardware.

04

Sanjavan *Ghodasara*

REALTIME · FRONTEND

Stevens MS ML · 8+ hackathon wins.